

Super sound

Sound energy travels through air at a speed of around 340 m (1,120 ft) per second. This speed is also known as Mach 1. If something travels faster than Mach 1, we say it is supersonic – that means it is travelling faster than sound.

Sonic boom

When an aircraft travels at the speed of sound, it creates a big noise called a sonic boom. It is made because the plane catches up with its own sound waves, squashing them together. Travelling at speeds near Mach 1 can also create a cone-shaped cloud called a vapour cone.

When an aircraft gets near supersonic speeds, the particles of air behind it spread out, which lowers the temperature.



The cone-shaped cloud is made of droplets of water that have condensed out of the air as the temperature fell.

Infrasound

The slower something vibrates, the lower the pitch of the sound it makes. Infrasound is sound that is too low for humans to hear. However some animals, such as elephants and whales, can make and hear these super-low noises.



Elephants use infrasound to communicate with each other.